OPEN SOURCE NIGHT

Nightshift

From Copilot to Autopilot

Engineering guardrails for AI coding agents

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- Skills
- Harness
- Routing
- Demonstration
- Evidence
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Joint Atlanta developer-community meetup

OPEN SOURCE

An inspectable engineering workflow

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A useful starting point

Read the specification and tests.

Understand what the change should do.

An inspectable contribution

Show the source diff and evidence.

Let another reviewer challenge the result.

> The repository exposes the instructions as well as the implementation.

AGENTS.md; commands/; agents/; scripts/

OVERVIEW

The harness keeps the workflow

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Entry point

A skill or CLI starts the work.

Canonical instructions

Stages and roles define the work.

Routing

Configuration selects the provider.

Controller

Evidence determines the next stage.

AGENTS.md; skills/nightshift/SKILL.md; routing.json; scripts/nightshift-pipeline.py

The skill is the entry point

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CODEX SKILL

```
$nightshift <ticket-ref>
```

The adapter loads the canonical stage instructions.

A ticket starts the engineering pipeline without separate calls for each stage.

SKILLS

The instructions stay in the repository

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commands/	Canonical stage instructions
agents/	Reusable role instructions
scripts/	Execution and validation
routing.json	Provider and model selection

AGENTS.md; commands/nightshift-product.md; agents/nightshift-spec-writer.md

Six stages retain their evidence

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Product

Specification and scope

2

Adversarial

Independent challenge

3

Implementation

Tests and source changes

4

Review

Independent code review

5

Drift

Specification alignment

6

QA

Final verification

commands/nightshift-{product,adversarial,implement,review,drift,qa}.md

A small task makes proof visible

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PUBLIC DURATION FORMATTER

```
format_seconds(0)    == "0:00"
format_seconds(125)  == "2:05"
format_seconds(3600) == "60:00"
```

Acceptance cases

Nonnegative integers format correctly.

Negative integers raise an error.

Booleans and other types raise an error.

Routing selects the worker

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Author path

Codex

`gpt-6.1-sol`

Product and Implementation

Independent review path

Claude

`sonnet`

Adversarial, Review, Drift and QA

> These are the recorded routes for this acceptance run.

routing.json; accepted controller receipts for issue 133

Admission happens before dispatch

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Provider capability

Required runtime and tools are available.

Local inventory

The configured local model exists before extraction starts.

Retained allowance

Existing call and active-time limits still apply.

Bounded context

Stage handoffs carry relevant files and findings.

scripts/nightshift-capability.sh; scripts/nightshift-ticket-budget.py; scripts/nightshift-handoff.py

DEMONSTRATION

The front end shows the evidence

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Routes, receipts, artifacts and retained attempts remain inspectable.

Actual dashboard capture of the verified public rehearsal

DEMONSTRATION

Watch Nightshift work

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PRESENTER TERMINAL · FROM THE COMMUNITY CHECKOUT

```
python3 scripts/nightshift-talk-check.py
```

Recheck six stage receipts, real tests and both proof gates.

Repeat the retained task with zero new generation calls.

Open the local dashboard ↗ · Demo setup and commands ↗

Presenter laptop: retained fixture and subscription login required. Cached repeat, not a fresh agent run.

DEMONSTRATION

RED and GREEN remain inspectable

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Before implementation

Retained RED evidence records expected test failures.

Frozen cases define the required behavior.

After implementation

The real tests pass.

Development and final proof gates both pass.

> The rejected receipt stays in history.

commands/nightshift-implement.md; docs/133/IMPLEMENTATION.md

EVIDENCE

Zero generation calls on repeat

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6	0	26
accepted stages	additional generation calls	unchanged call reservations

Receipt hashes and attempt history stayed unchanged.

Current bindings allow the controller to reuse accepted work.

docs/133/VALIDATION.json; retained repeat-validation.json

Verification has a defined scope

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Observed

One public task passed all six stages.

Both proof gates and repeat reuse passed.

Ten CI shards and dashboard checks passed.

Still outside this evidence

Reliability across arbitrary projects

Actual provider billed charges

Hermes or Jev accuracy and savings

docs/133/VALIDATION.json; GitHub Actions run 38600812302

NIGHTSHIFT BUILDS NIGHTSHIFT

We pointed it at itself

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Build

A control-room UI for Nightshift.

2

Challenge

Review catches a hidden continuation control.

3

Repair

Restore access. Tests and re-review pass.

> It caught a regression—and exposed its own workflow bug.

Merged after operator approval.
The original run stopped at its work allowance.

#138 · Real run and evidence ↗ #139 · Human acceptance sent to code repair ↗ #140 · Operator-directed merge ↗

DEEP DIVE

Human checkpoints

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Declared operating mode

Supervised planning waits for approval.

Autonomous planning records its plan and continues.

Explicit boundaries

Production deployment requires confirmation.

Permanent removal requires confirmation.

> Passing a test does not grant new authority.

AGENTS.md; commands/nightshift-implement.md

A contribution people can review

A clear behavior, a focused change and evidence someone else can inspect.

github.com/doctor-ew/nightshift-community

Questions and discussion

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APPENDIX

Testing the retained demo

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FROM THE COMMUNITY CHECKOUT

```
python3 scripts/nightshift-talk-check.py
```

The checker validates identity, receipts, proof and real tests.

The repeat check refuses new provider generation.

scripts/nightshift-talk-check.py; docs/talks/TESTING.md

DEEP DIVE

SOLID and ACID in review

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SOLID

Responsibilities and interface contracts

Dependencies and extension points

ACID

Validation and consistent state

Isolation and recovery for state changes

> These are review lenses. Findings still need evidence.

commands/nightshift-review.md